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A pairwise likelihood approach to estimation in multilevel probit models

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Abstract

A pairwise likelihood (PL) estimation procedure is examined in multilevel models with binary responses and probit link. The PL is obtained as the product of bivariate likelihoods for within-cluster pairs of observations. The resulting estimator still enjoys desirable asymptotic properties such as consistency and asymptotic normality. Therefore, with this approach a compromise between computational burden and loss of efficiency is sought. A simulation study was conducted to compare PL with second-order penalized quasi-likelihood (PQL2) and maximum (marginal) likelihood (ML) estimation methods. The loss of efficiency of the PL estimator is found to be generally moderate. Also, PL tends to show more robustness against convergence problems than PQL2.

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1. Introduction

Multilevel modeling ([Goldstein, 1995](#)) refers to a class of statistical techniques for the analysis of data that have a hierarchical structure, i.e., that were collected in groups or “clusters”, possibly at several levels of a hierarchy. The popularity of multilevel modeling techniques has been growing over the last two decades as this type of data naturally arises in many scientific disciplines. This has certainly been rendered possible with the widespread availability of modern computing power.

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The distinguishing feature of clustered data is that observations within a cluster tend to be more alike than observations from different clusters or, stated otherwise, they are correlated. Heterogeneity between clusters therefore introduces an additional source of variation which complicates analysis. When this extra variation cannot be explained by measured covariates, we require statistical analysis methods which acknowledge explicitly for the clustering in the data.

For continuous outcomes, the theory and implementation of the multilevel linear model are greatly simplified owing to the elegant properties of the multivariate normal distribution. When the outcome variable is categorical, a first issue arises which is the lack of a discrete analogue to the multivariate normal distribution. One also needs to distinguish between different kinds of models such as marginal models wherein the first, second and possibly higher moments are modeled separately in terms of covariates (e.g. [Liang and Zeger, 1986](#)), or random-effects models (also commonly named generalized linear mixed models (GLMM)), wherein dependencies among observations arise by introducing shared random variables in the model. [Agresti et al. \(2000\)](#) provide an excellent review of random-effects modeling of categorical response data with social science applications.

A problem inherent in GLMMs is that the marginal likelihood function obtained after integrating over random effects nearly always involves intractable integrals. A number of approximate procedures have therefore been proposed in the literature to avoid the need for numerical integration in GLMMs. [Breslow and Clayton \(1993\)](#), for instance, exploit the penalized quasi-likelihood (PQL) estimator by applying Laplace's method for integral approximation. They also consider marginal quasi-likelihood (MQL), a name they give to a procedure previously proposed by [Goldstein \(1991\)](#). These two approaches entail iterative fitting of linear models based on first-order Taylor expansions of the mean function about the current estimated fixed part predictor (MQL) or the current predicted value (PQL).

Based on simulated data, [Rodríguez and Goldman \(1995\)](#) demonstrate that these approximate procedures may be seriously biased when applied to binary response data. Their simulations reveal that both fixed effects and variance components may suffer from substantial, if not severe, attenuation bias under certain circumstances. [Goldstein and Rasbash \(1996\)](#) show that including a second-order term in the PQL expansion (PQL2) essentially eliminates biases described by Rodríguez and Goldman. Other authors have also advised the introduction of bias-correction terms ([Breslow and Lin, 1995](#)). It should be noted that the PQL algorithm can be numerically unstable and it is not so uncommon that it fails to converge in practical applications. This situation tends to worsen when second-order terms are added or when the model becomes more complicated ([Renard et al., 2002](#)).

In this paper, a pairwise likelihood approach to fit multilevel models to binary data is examined. Pairwise likelihood is a special example of what is called pseudo-likelihood, first proposed by [Besag \(1975\)](#) and also termed composite likelihood by [Lindsay \(1988\)](#). The motivation behind pseudo-likelihood estimation is to replace the likelihood by a function that is easier to evaluate, and hence to maximize. The function in question is a product of conditional or marginal densities. Thus, the main feature of a pseudo-likelihood function is that it is composed of (pieces of) likelihoods and this can

be exploited to prove general results about the consistency and asymptotic normality of pseudo-likelihood estimators

Examples of pairwise likelihood estimation to multivariate correlated binary data were proposed by [Le Cessie and Van Houwelingen \(1994\)](#), [Geys et al. \(1997, 1998\)](#), and [Kuk and Nott \(2000\)](#). These authors, however, have focused on population-averaged models, as opposed to the cluster-specific modeling approach endorsed in the present paper. Another related work is that of [Heagerty and Lele \(1998\)](#) who used pairwise likelihood with binary spatial data.

An application of the methodology to real data will not be presented here. Instead, we refer to the paper by [Renard et al. \(2002\)](#), which deals with the validation of surrogate endpoints in a meta-analytic setting when both endpoints are binary. This approach to surrogate marker validation assumes a joint model for the surrogate and the true endpoints and entails fitting of a three-level probit model with four-dimensional normal random effects. In fact, the interest of the present paper originated in this former work to further evaluate performances of the PL estimator in simpler settings where maximum (marginal) likelihood (ML) and PQL2 can be easily employed. To that end, after describing the PL estimation procedure in Section 2, we report results of a simulation study in Section 3 to compare PL to ML and PQL2 using a two-level model with random intercept only or with random intercept and slope. We then conclude the paper by a more extensive discussion of the relative merits of these estimation procedures.

2. Pairwise likelihood in the multilevel probit model

2.1. The multilevel probit model

For simplicity, we assume a two-level data structure. This would be the case when level 1 units (e.g. repeated measurements in a longitudinal study or offsprings in a teratologic study) are nested within level 2 units (i.e., subjects in the longitudinal context or litters in the teratologic application). The procedure described hereafter can, in principle, be applied to general multilevel structures, although practical limitations at the computational level arise (see Section 4 for further discussion).

As it will become apparent, a probit link must be postulated for full computational gains. The logit link, on the other hand, has been quite popular in marginal models because parameters are interpretable in terms of (log) odds ratios. By extension, it has been mostly studied in multilevel generalized linear models as well. In addition to the fact that both links tend to provide similar model fits in practice whatever modeling approach is followed, we do believe that the logit link function is less appealing with random-effects modeling because of the cluster-specific interpretation of the parameters. Moreover, the probit link is more convenient to make population-averaged inference since unconditional (or marginal) probabilities are simple probit functions.

We now introduce the multilevel probit model from a latent variable perspective. Let $\mathbf{Y}_i = (Y_{ij} : j = 1, \dots, n_i)^T$ denote the vector of binary measurements on unit i ($i = 1, \dots, N$). We assume that each Y_{ij} takes on a value of 0 or 1. The model for the probability that, conditionally on a set of random effects $\mathbf{b}_i$, a positive response be

observed on the j th unit in the i th cluster is specified as

$$\Phi^{-1}(P[Y_{ij} = 1 | \mathbf{b}_i]) = \mathbf{x}_{ij}^T \boldsymbol{\beta} + \mathbf{z}_{ij}^T \mathbf{b}_i, \quad (1)$$

where $\mathbf{x}_{ij}$ is a vector of covariates having fixed effects $\boldsymbol{\beta}$ and $\mathbf{z}_{ij}$ denotes a vector of covariates, possibly overlapping with $\mathbf{x}_{ij}$, having random effects $\mathbf{b}_i \sim N(0, \Sigma_b)$ of dimension q .

We posit the existence of a latent variable $\tilde{Y}_{ij}$ that is continuously distributed and related to the actual response through a certain threshold. In the context of iid binary data, this approach motivates a wide class of models, of which the standard logistic and probit regression models are special cases (Cox and Snell, 1989). We assume that the observed binary response is actually obtained by dichotomizing an unobserved continuous variable, hence the need for a certain threshold or ‘cut-off’ value. This threshold can be chosen to be 0 without loss of generality, provided an intercept term is included in the model. In other words, it is assumed that a positive response is recorded ($Y_{ij} = 1$) if $\tilde{Y}_{ij} > 0$ and a negative response ($Y_{ij} = 0$) otherwise. If we further assume that $\tilde{Y}_{ij}$ is normally distributed, then the random-effects regression model for the latent response variable can be written as follows:

$$\tilde{Y}_{ij} = \mathbf{x}_{ij}^T \boldsymbol{\beta} + \mathbf{z}_{ij}^T \mathbf{b}_i + \tilde{\epsilon}_{ij}, \quad (2)$$

where the residual error terms $\tilde{\epsilon}_{ij}$ are assumed to be normally distributed with mean zero and variance σ_e^2 .

We note that the parameter σ_e^2 is not identifiable in the model. For identifiability, its value is fixed, without loss of generality, to 1. With this choice, the derived model for the binary response variable Y_{ij} is exactly (1).

Likelihood estimation proceeds by maximizing the marginal distribution obtained by integrating over the random effects. To further simplify ideas, we assume that $Z_{ij} = 1$ for all i and j in the remainder of this section, and we will write σ_b^2 for Σ_b^2 in this case. This corresponds to a random-intercept model, with b_i representing the random departure of the i th cluster. The contribution of the i th cluster to the likelihood function for this model can then be written

$$\ell_i(\boldsymbol{\beta}, \sigma_b^2) = \int_{-\infty}^{+\infty} \prod_{j=1}^{n_i} P[Y_{ij} = 1 | b_i] \phi(b_i; \sigma_b^2) db_i, \quad (3)$$

where $\phi(b_i; \sigma_b^2)$ denotes the density function of the normal distribution with mean 0 and variance σ_b^2 . In this example, Gauss–Hermite quadrature (Anderson and Aitkin, 1985) can be easily used to approximate the integral in (3). This method, however, suffers from the curse of dimensionality. In addition, no simple and effective error bounds are available for multidimensional integrals approximated by quadrature rules. For these reasons ML estimation in GLMMs has been somewhat hampered in the past, being limited to simplistic models with low-dimensional random effects, and there has been a need for alternative, performant integration methods to overcome computational limitations. Among those, Monte Carlo-based methods have received a great deal of attention due to parallel developments in Bayesian statistics. For example, Monte Carlo expectation maximization (EM) and Monte Carlo Newton Raphson algorithms (McCulloch, 1994, 1997) were proposed to obtain ML estimates. Alternatively, the

generalized linear random-effects model can be cast in a fully Bayesian framework (see e.g. [Zeger and Karim, 1991](#)). A drawback of these methods is that they are computationally intensive in general.

2.2. Pairwise likelihood

When maximization of the likelihood is awkward, an alternative is to use a pseudo-likelihood obtained by replacing the joint likelihood by any suitable product of conditional or marginal densities. We refer to [Arnold and Strauss \(1991\)](#) for more details and examples. In this paper, attention is restricted to products of pairwise likelihoods. More formally, the basic contribution of the i th cluster to the log PL can be written

$$p\ell_i(\boldsymbol{\beta}, \sigma_b^2) = \sum_{j=1}^{n_i} \sum_{k>j} \sum_{l,m=0}^1 \delta_{ijklm} \log P[Y_{ij} = l, Y_{ik} = m] \quad (4)$$

with

$$\delta_{ijklm} = \begin{cases} 1 & \text{if } Y_{ij} = l \text{ and } Y_{ik} = m, \\ 0 & \text{otherwise.} \end{cases}$$

We emphasize that pairwise probabilities in (4) are marginal, not conditional, probabilities. These marginal pairwise probabilities are straightforward to calculate in terms of univariate and bivariate probits. For example, using the latent variable formulation of the model it is easy to show that

$$P[Y_{ij} = 1, Y_{ik} = 1] = \Phi_2(\xi_{ij}, \xi_{ik}; \rho_{ijk}),$$

where the function $\Phi_2(x, y; \rho)$ denotes the standardized bivariate normal distribution function with correlation coefficient ρ , and where we define

$$\xi_{ij} = \frac{\mathbf{x}_{ij}^T \boldsymbol{\beta}}{\sqrt{\text{var}[\tilde{y}_{ij}]}}$$

and

$$\rho_{ijk} = \text{corr}[\tilde{y}_{ij}, \tilde{y}_{ik}].$$

In this expression, $\text{var}[\tilde{Y}_{ij}]$, $\text{var}[\tilde{Y}_{ik}]$ and $\rho_{ijk} = \text{corr}[\tilde{Y}_{ij}, \tilde{Y}_{ik}]$ are obtained by selecting the appropriate 2×2 submatrix of the covariance matrix $V_i = Z_i \Sigma_b Z_i^T + I_{n_i}$, where $Z_i = (\mathbf{z}_{i1}, \dots, \mathbf{z}_{in_i})^T$ and I_{n_i} is the identity matrix of dimension n_i . The function $\Phi_2(x, y; \rho)$ denotes the standardized bivariate normal distribution function with correlation coefficient ρ .

The PL estimators $(\tilde{\boldsymbol{\beta}}, \tilde{\sigma}_b^2)$ are obtained by maximizing the function

$$\log PL(\boldsymbol{\beta}, \sigma_b^2) = \sum_{i=1}^N p\ell_i(\boldsymbol{\beta}, \sigma_b^2). \quad (5)$$

General results on the consistency and asymptotic normality of PL estimators can be derived along the lines of classical proofs for maximum likelihood estimators

(Arnold and Strauss, 1991). If $\tilde{\theta}$ denotes the PL estimator of $\theta = (\beta^T, \sigma_b^2)^T$, then the asymptotic covariance matrix of $\tilde{\theta}$ can be approximated by the “sandwich estimator” $J^{-1}KJ^{-1}$, where

$$J = -\sum_{i=1}^N \frac{\partial^2 p\ell_i(\tilde{\theta})}{\partial \theta \partial \theta^T}$$

and

$$K = \sum_{i=1}^N \frac{\partial p\ell_i(\tilde{\theta})}{\partial \theta} \frac{\partial p\ell_i(\tilde{\theta})}{\partial \theta^T}.$$

Alternatively, an estimator of

$$E \left[-\frac{\partial^2 \log PL}{\partial \theta \partial \theta^T} \right],$$

which does not require evaluation of second-order derivatives is given by

$$J = -\sum_{i=1}^N \sum_{j=1}^{n_i} \sum_{k>j} \frac{\partial \log P(y_{ij}, y_{ik})}{\partial \theta} \frac{\partial \log P(y_{ij}, y_{ik})}{\partial \theta^T}.$$

As can be anticipated, the PL estimator will be, in general, less efficient than the maximum likelihood estimator since it relies on a restricted amount of information. Therefore, with this approach a compromise between computational burden and loss of efficiency is sought.

2.3. Weighted pairwise likelihood

Expression (4) shows how the log pairwise likelihood can be constructed. Interestingly, each response Y_{ij} occurs $(n_i - 1)$ times in this expression. In consequence, when cluster sizes are unequal, an observation coming from a large cluster will contribute more information to the log pairwise likelihood than one taken from a small cluster and the issue arises as to whether one should weight or not each term in the log pairwise likelihood.

Le Cessie and Van Houwelingen (1994) argued that $p\ell_i$ should be inversely weighted by a factor $(n_i - 1)$, which led them to define the weighted (log) pairwise likelihood (WPL):

$$\log PL^*(\beta, \sigma_b^2) = \sum_{i=1}^N \frac{p\ell_i(\beta, \sigma_b^2)}{n_i - 1}. \quad (6)$$

Their argument is that if the observations are in fact independent, the contribution of each observation is counted $(n_i - 1)$ times in $p\ell_i$ and so observations in the large clusters are given more weight than observations in the small clusters whereas they should be treated equally under independence. Therefore, the log WPL reduces to the log likelihood under independence and can be expected to be nearly efficient under weak dependence.

In marginal models with exchangeable correlation structure, [Geys et al. \(1998\)](#) show that PL and GEE2 approaches have strong common bases. More specifically, the set of GEE2 equations for estimating marginal parameters is closely related to the corresponding set of WPL equations, whereas the set of GEE2 equations for estimating association parameters is closely related to the corresponding set of PL equations. The authors conclude that when main interest lies in the marginal parameters, WPL should be preferred whereas if main interest lies in the association parameters, use of PL should be recommended.

[Kuk and Nott \(2000\)](#) support this conclusion using an argument based on the theory of optimal estimating functions. They show that for a simple model with marginal probability π (no covariates) and common pairwise correlation ρ , the optimally weighted score equations are

$$\sum_{i=1}^N \frac{1}{n_i - 1} \frac{\partial p\ell_i^{(m)}}{\partial \pi} = 0 \quad (7)$$

and

$$\sum_{i=1}^N \frac{\partial p\ell_i^{(m)}}{\partial \rho} = 0, \quad (8)$$

where $p\ell_i^{(m)}$ is the log pairwise likelihood from the marginal model for cluster i . In other words, derivatives with respect to the marginal parameter π should be weighted but not those with respect to the association parameter ρ . Hence to solve this pair of equations, one should alternate between maximizing WPL with respect to π for a given ρ and maximizing PL with respect to ρ for a given π .

We shall now apply the argument of [Kuk and Nott \(2000\)](#) to the multilevel probit model of interest here. To this end, we consider a model with a single intercept term β (no covariates) and a random intercept with variance $\theta = \sigma_b^2$.

The argument of [Kuk and Nott \(2000\)](#) proceeds by noting that the optimal weighting of $p\ell_i$ in the score equation is given by

$$\sum_{i=1}^N J_i K_i^{-1} \frac{\partial p\ell_i}{\partial (\beta, \theta)^T} = 0,$$

where

$$J_i = E \left[- \frac{\partial^2 p\ell_i}{\partial (\beta, \theta)^T \partial (\beta, \theta)} \right]$$

and

$$K_i = \text{cov}[p\ell_i] = E \left[\frac{\partial p\ell_i}{\partial (\beta, \theta)^T} \frac{\partial p\ell_i}{\partial (\beta, \theta)} \right].$$

They then evaluate J_i and K_i under the simplifying assumption of independence. Following the same lines of reasoning, we find after some straightforward but tedious calculations:

$$J_i K_i^{-1} = \begin{pmatrix} \frac{1}{n_i - 1} & 0 \\ \frac{(n_i - 2)\beta}{n_i - 1} & 1 \end{pmatrix}.$$

Table 1

Asymptotic efficiency of PL versus WPL in the random-intercept model. Cell entries are asymptotic relative efficiencies (percentages) for $\hat{\beta}$ (first row) and $\hat{\theta}$ (second row)

β	θ					
	0	0.25	0.5	1	2	4
0	94.0	91.0	89.8	88.8	88.0	87.5
	107.8	100.1	97.4	95.1	93.4	92.3
1	97.5	94.3	92.7	91.1	89.7	88.7
	107.8	100.8	97.8	95.2	93.4	92.3
2	105.3	101.2	98.3	95.2	92.8	91.0
	107.8	102.9	99.2	95.7	93.5	92.3
3	107.6	105.2	101.6	97.4	94.4	92.4
	107.8	105.5	101.5	96.8	93.8	92.3

In comparison, [Kuk and Nott \(2000\)](#) found

$$J_i K_i^{-1} = \begin{pmatrix} \frac{1}{n_i - 1} & 0 \\ 0 & 1 \end{pmatrix},$$

which led them to the pair of Eqs. (7) and (8).

From the above argument we obtain a weighted equation for the fixed parameter but a weighted sum of pseudo-score functions for the random parameter. It can be noted, however, that the weights will not be strongly dependent on cluster size—at least when clusters are large since $(n_j - 2)/(n_j - 1) \simeq 1$ in this case—but the resulting equation will depend on the fixed parameters, which prevents us from drawing the same conclusions as [Kuk and Nott \(2000\)](#).

To supplement this result and gain further insight into the problem, we report some asymptotic efficiency calculations to compare the weighted and unweighted PL estimators. We consider the same model as in the above discussion, with $\beta = 0, 1, 2, 3$ and $\theta = 0, 0.25, 0.5, 1, 2, 4$. Clusters are either of size 3, 4, or 5 and assumed each to occur with equal probability. Table 1 reports asymptotic efficiencies of PL relative to WPL for each combination of parameter values. When there is none or little intra-cluster correlation (i.e., θ is close to 0), the PL estimator appears to be more efficient for the random parameter but for the fixed parameter the conclusion depends on its true assumed value. If the success probability evaluated at the mean of the random-effect distribution is large (i.e., β is small), the WPL estimator is more efficient than the PL estimator whereas the reverse holds true for more extreme success probabilities. As the association among responses becomes stronger, the WPL estimator tends to be more efficient than the PL estimator for both parameters. In conclusion, none of the PL or WPL estimators is uniformly more efficient than the other and losses of efficiency between the two estimators are modest (less than 10%). In view of Table 1, the WPL estimator tends to perform slightly better than the PL estimator overall, especially

when $\theta > 0.5$, that is, for values of θ that we have found to be relatively common in practice.

2.4. Practical implementation

To conclude this section, we briefly comment on the practical implementation of the algorithm. To ensure that Σ_b be positive-definite and to improve the convergence properties of the algorithm, a Cholesky decomposition $D^T D = \Sigma_b$ is used and the PL function maximized with respect to the elements of the Cholesky factor. The algorithm was implemented in SAS IML (SAS Institute Inc., 1995) and maximization of the log PL performed using the NLPDD (Double-Dogleg) optimization routine (SAS Institute Inc., 1995). This optimization procedure requires only function and gradient calls which are less expensive to evaluate than second-order derivatives. The final hessian matrix was computed using numerical second-order derivatives by forward difference approximations. Of course, the more expensive but accurate central difference approximations could be used instead.

3. Simulation study

A simulation study was conducted to further evaluate the performances of the PL estimator in finite samples and compare it with the ML and PQL2 estimators in relatively simple two-level models for binary responses. Of particular interest are the loss of efficiency induced by PL and the overall bias/variance trade-off.

The underlying model from which data were simulated was taken to be

$$\Phi^{-1}(P[y_{ij} = 1 | \mathbf{b}_i]) = \beta_0 + \beta_1 x_{ij} + \mathbf{z}_{ij}^T \mathbf{b}_i \quad (9)$$

with $\beta_0 = 0.5$ and $\beta_1 = 1$. The results reported here are for a covariate x with values generated from a uniform distribution over $[-1, 1]$ for each observation in a cluster. The case of a cluster-specific covariate was also considered and conclusions are briefly discussed below. Model (9) assumed either a random intercept ($\mathbf{z}_{ij} = 1$) or a random intercept and slope ($\mathbf{z}_{ij} = (1, x_{ij})^T$). The random effects were generated from independent normal distributions with identical variances (0.5 or 1). To form the binary outcome y_{ij} , we further generated individual random components $N(0, 1)$ so that the binary outcome was set to 1 when the linear predictor plus the sum of the random effects and the individual random component was greater than 0. The number of clusters was fixed to either 20 or 50, and all clusters had sizes taken from a uniform distribution on $[10, 30]$. The results were based on 500 replicates under each scenario.

For each generated data set, parameters were estimated using ML, PQL2 and PL procedures. For ML, adaptive Gaussian quadrature with 10 quadrature nodes was utilized to evaluate the marginal likelihood function. The Double-Dogleg routine was employed to maximize the marginal likelihood function and the sample information matrix was estimated using numerical second-order derivatives (with forward difference

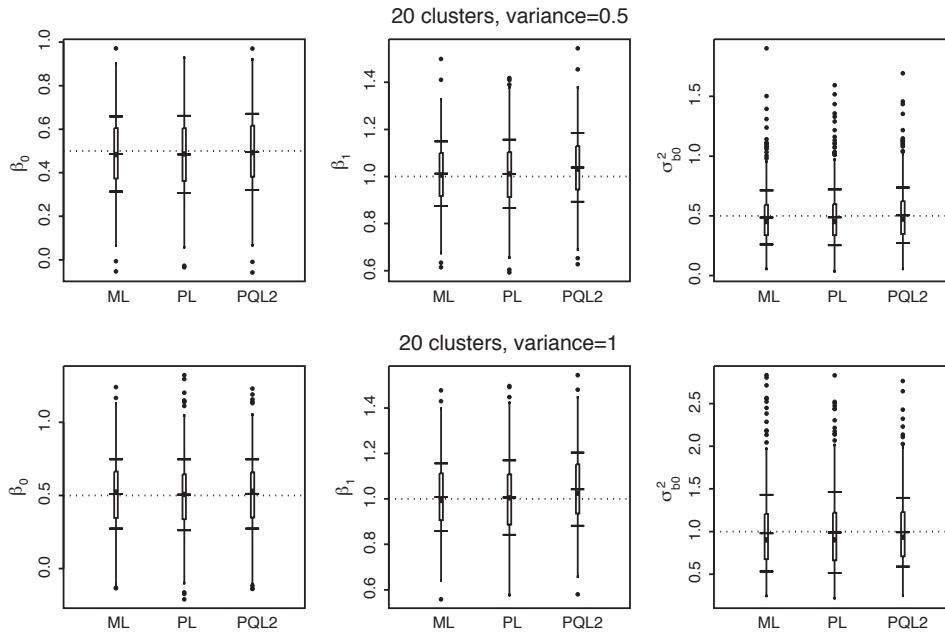


Fig. 1. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (9) with random intercept $\sim N(0, \sigma_{b0}^2)$. Top panel: 20 clusters with $\sigma_{b0}^2 = 0.5$; Bottom panel: 20 clusters with $\sigma_{b0}^2 = 1$.

approximations) upon convergence. To obtain PQL2 estimates, the approach of Goldstein and Rasbash (1996), and its implementation in MLwiN (Goldstein et al., 1998), were followed. Finally, the PL procedure was implemented as outlined in Section 2. Results reported hereafter are for the weighted estimator as it was found more efficient than the unweighted estimator under the chosen simulation settings. Note that for ML and PL, an absolute gradient convergence criterion of 0.001 was specified, whereas for PQL2 a relative parameter criterion of 0.001 was used.

Results for the random-intercept model are summarized in Figs. 1 and 2. These figures present box and whisker plots of the simulated estimates for each parameter. The whiskers extend from the interquartile range (the box) as far as the data extend to a distance not exceeding 1.5 interquartile ranges. The thick mark in the box represents the median. Also included as horizontal lines are the mean ± 1 root mean-square error (MSE). Note that in these figures and subsequent ones, we have occasionally removed a few extreme values from the plots (but not from the calculations of summary statistics) for better readability.

In both figures estimates of the fixed parameters show no systematic bias, except for the slope parameter which seems to be slightly biased upward under PQL2. Estimates of σ_{b0}^2 were located, on average, on the true value of the parameter and their distribution is, as can be expected, slightly skewed to the right, this being more pronounced with smaller sample sizes (Fig. 1). When comparing MSEs of β_0 and β_1 ,

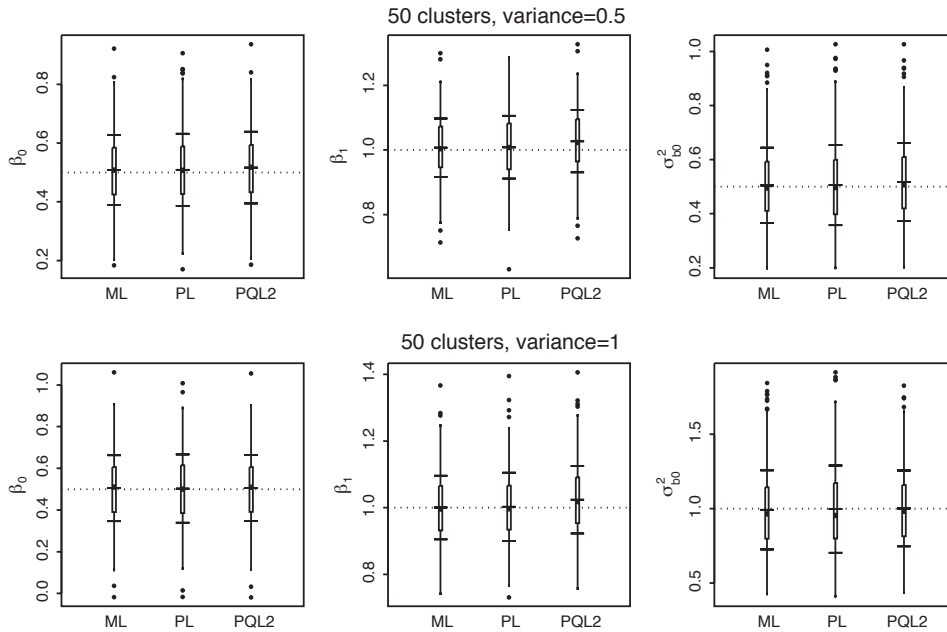


Fig. 2. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (9) with random intercept $\sim N(0, \sigma_{b0}^2)$. Top panel: 50 clusters with $\sigma_{b0}^2 = 0.5$; Bottom panel: 50 clusters with $\sigma_{b0}^2 = 1$.

there is little difference across all three estimation methods, with a slight advantage to ML. For σ_{b0}^2 , MSEs were also smaller with ML when $\sigma_{b0}^2 = 0.5$ but were, surprisingly, in favor of PQL2 when $\sigma_{b0}^2 = 1$. When looking at the relative efficiency of the PL estimator to the ML estimator (which was estimated as the ratio of the variance of the ML parameter estimates over that of the PL estimates), efficiency losses were around 10% (ranging from 5% to 18%) for all parameters across the four simulation settings.

We found the good performance of the PQL2 estimator when $\sigma_{b0}^2 = 1$ quite surprising as this goes against the preconceived idea that PQL2 is biased downwards for large variance components. It should be stressed, however, that PQL2 has been studied mostly in models with logit link, which suggests that the link specification may be an important consideration when fitting multilevel models to binary data using the PQL2 algorithm. To further investigate this hypothesis, 1000 data sets of 50 clusters were generated from the random-intercept model, with the value of the variance parameter equal to 1. This was done using logit and probit links and results are summarized in Table 2. Note that the simulation setting is comparable to the one used by Goldstein (1995, p. 100). Results for the logit link are indeed very close to those reported by Goldstein and show the downward bias present in σ_{b0}^2 , with the values of the fixed parameters close to their true value. For the probit link the bias in the variance parameter is essentially eliminated, but there apparently is a bias transfer to the slope parameter,

Table 2

Simulations results (1000 replicates) to compare PQL2 estimates in the random-intercept model with logit and probit link. Means are reported with Monte Carlo error given between parentheses

Parameter	Probit	Logit
β_0	0.506 (0.005)	0.499 (0.005)
β_1	1.028 (0.003)	1.008 (0.004)
σ_{u0}^2	0.991 (0.008)	0.963 (0.009)

now with upward tendency. The reason behind this is not entirely clear. It can be observed, however, that the PQL algorithm entails iterative fitting of linear models that assume normality of the level 1 residuals. The latter assumption seems, in fact, better satisfied with the probit link in this particular situation. This, in turn, might lead to improved estimation of the variance parameter. We will see below, however, that this is not always so.

Results for the random-intercept-and-slope model are summarized in Figs. 3 and 4. The three estimation procedures yielded, on average, similar results for the fixed parameters β_0 and β_1 . No clear bias tendency could be detected in PQL2 estimates under this scenario. MSEs for β_0 were comparable under ML and PQL2 when $\sigma_{b00}^2 = \sigma_{b11}^2 = 0.5$ and were smaller under PQL2 when $\sigma_{b00}^2 = \sigma_{b11}^2 = 1$. Variance parameters exhibit substantially more variability here. In Fig. 3, the distribution of ML and PL estimates for σ_{b00}^2 and σ_{b11}^2 is shifted towards larger values, whereas the opposite happens with PQL2. With larger sample sizes (Fig. 4), ML and PL estimators were located, on average, on their true values but this was less clear with PQL2 (see bottom panel). Interestingly, though, MSEs of σ_{b00}^2 and σ_{b11}^2 were lower with PQL2. The covariance parameter σ_{b01} was, on average, correctly estimated with all three estimation methods. Finally, if we look at the relative efficiency of PL to ML, the loss was typically less than 15% for the fixed-effects parameters and σ_{b00}^2 , and about 40% for σ_{b01} and σ_{b11}^2 .

When the covariate x was cluster-specific (i.e., taking the same value for all observations within a cluster), conclusions remained essentially unchanged in the random-intercept model. In the random-intercept-and-slope model, convergence was an issue with all three procedures. With the smaller sample sizes (20 clusters), convergence rates were about 60% with ML and PL and dropped to less than 30% with PQL2. Fixed-effects parameters were reasonably well estimated but variance parameters were relatively biased (downwards for σ_{b00}^2 and upwards for σ_{b11}^2) and their distribution markedly skewed with all three procedures. With larger sample sizes, convergence rates were about 80% with ML and PL and about 50% with PQL2. The bias in the variance parameters was still present albeit to a lesser extent. The efficiency loss of the PL estimator compared to the ML estimator was moderate ($< 20\%$) in both settings.

To further evaluate the performance of PL in data sets with clusters of smaller size, an additional set of simulations was performed. The intent was to match a longitudinal

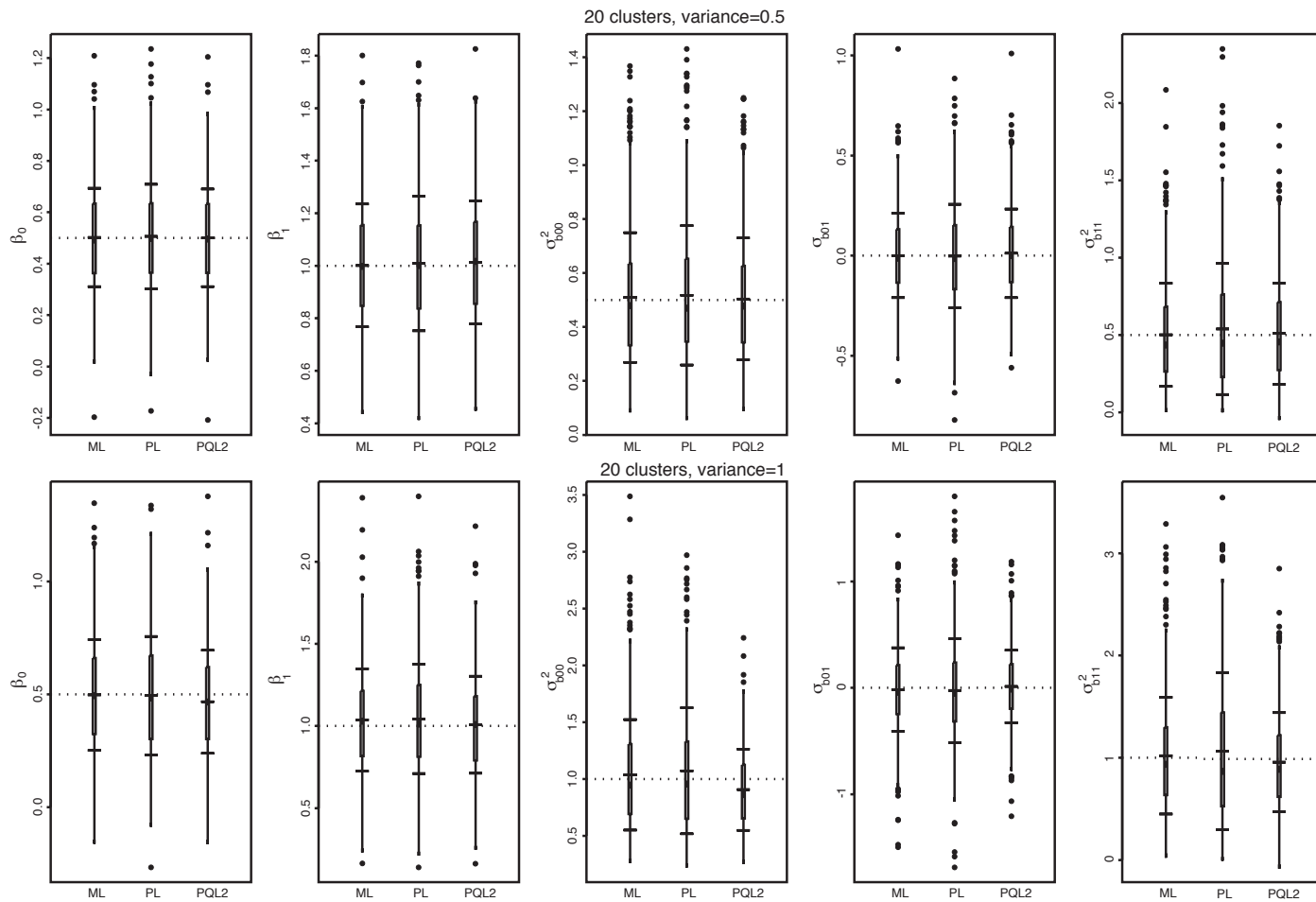


Fig. 3. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (9) with random intercept and random slope $\sim N(0, \Sigma_b)$. Top panel: 20 clusters with $\sigma_{b00}^2 = 0.5$, $\sigma_{b01} = 0$ and $\sigma_{b11}^2 = 0.5$; Bottom panel: 20 clusters with $\sigma_{b00}^2 = 1$, $\sigma_{b01} = 0$ and $\sigma_{b11}^2 = 1$.

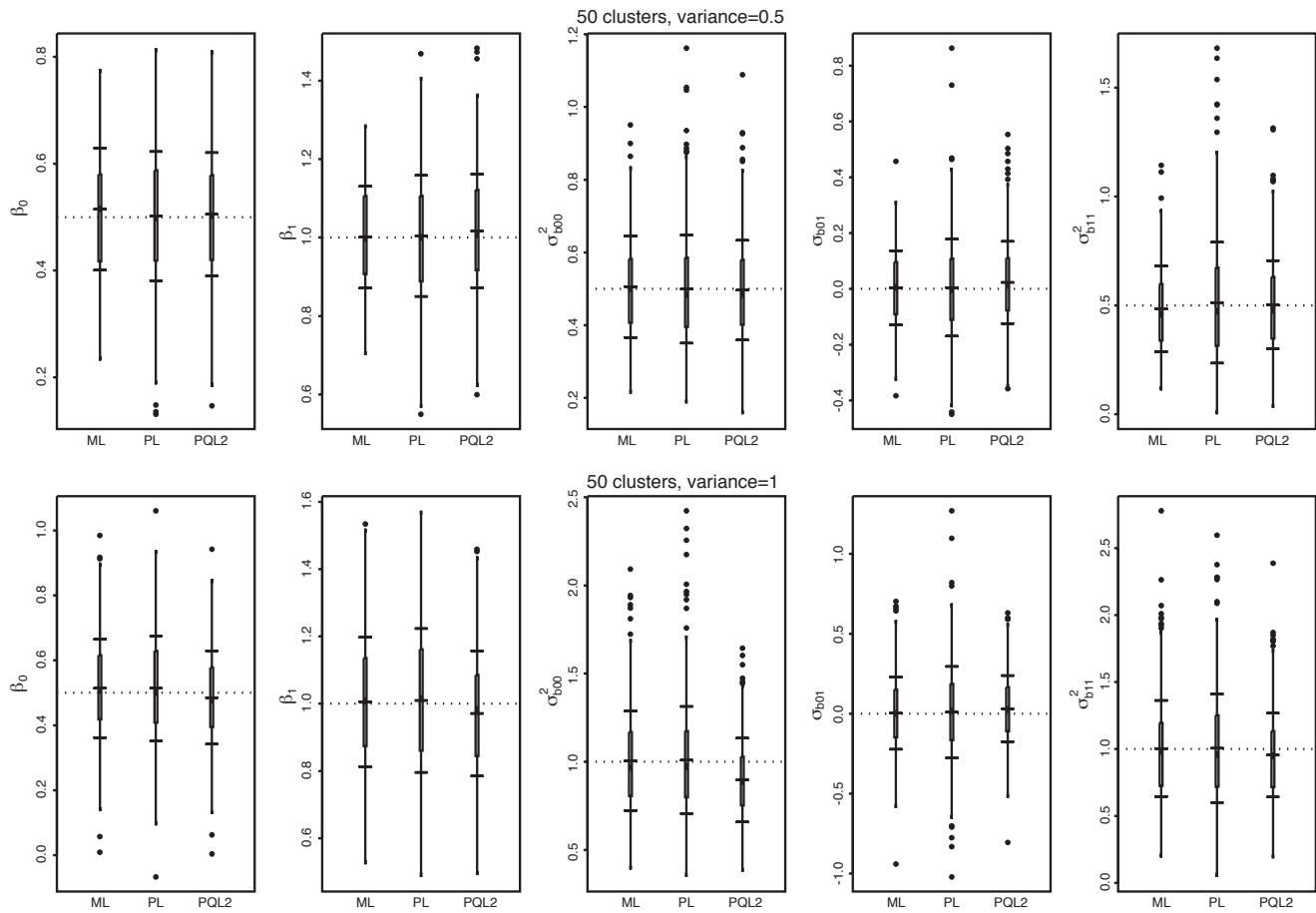


Fig. 4. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (9) with random intercept and random slope $\sim N(0, \Sigma_b)$. Top panel: 50 clusters with $\sigma_{b00}^2 = 0.5$, $\sigma_{b01} = 0$ and $\sigma_{b11}^2 = 0.5$; Bottom panel: 50 clusters with $\sigma_{b00}^2 = 1$, $\sigma_{b01} = 0$ and $\sigma_{b11}^2 = 1$.

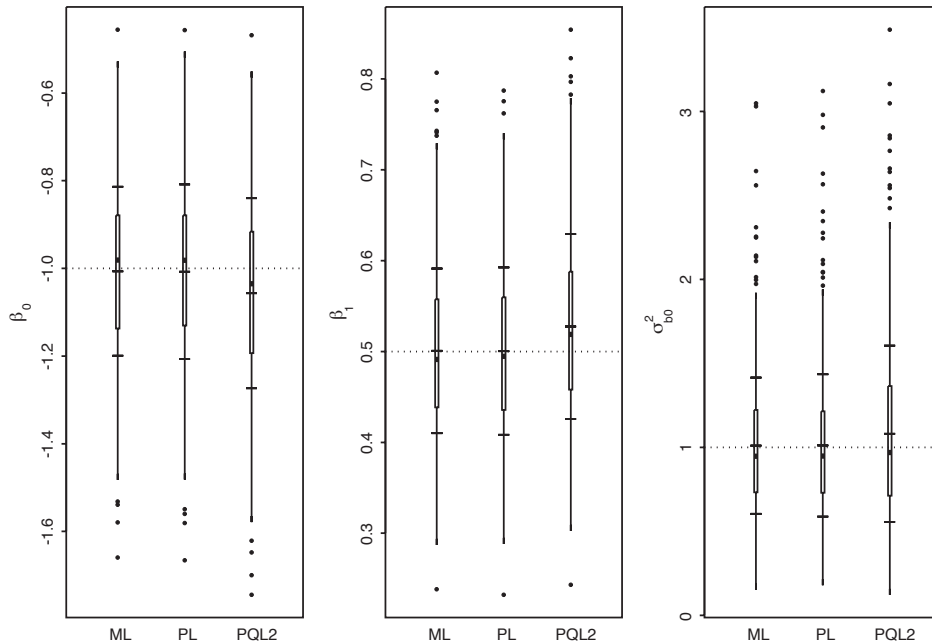


Fig. 5. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (10) with random intercept.

study design, so these simulations were based on the following model:

$$\Phi^{-1}(P[y_{ij} = 1 | \mathbf{b}_i]) = \beta_0 + \beta_1 t_{ij} + \mathbf{z}_{ij}^T \mathbf{b}_i \quad (10)$$

with $\beta_0 = -1$, $\beta_1 = 0.5$ and $t_{ij} = 0, \dots, 4$. The random effects had variance equal to 1. The number of subjects was taken to be 100, each subject having between 1 and 5 observations (randomly determined).

Results for the random-intercept model are summarized in Fig. 5. ML and PL procedures behave comparably, with a small loss of efficiency for PL compared to ML ($< 10\%$). Interestingly, PQL2 seemed to perform worse under this scenario: all parameters appear to be biased and to exhibit more variability.

For the random-intercept-and-slope model (Fig. 6), convergence difficulties were more frequent. With PQL2, a solution could be obtained in approximately 75% of the cases. With ML and PL procedures, convergence was easier to achieve ($\approx 95\%$ of the cases) although some solutions had unusually large values for some of the parameters, especially (co)variance parameters. After deleting these cases, we were left with about 90% of the observations. Note that the convergence criterion was lowered in order to detect solutions converging towards a non positive-definite covariance matrix Σ_b , which partly caused the aforementioned problem. Fig. 6 shows that ML and PL behave comparably, with up to 30% loss of efficiency for PL compared to ML. Even though

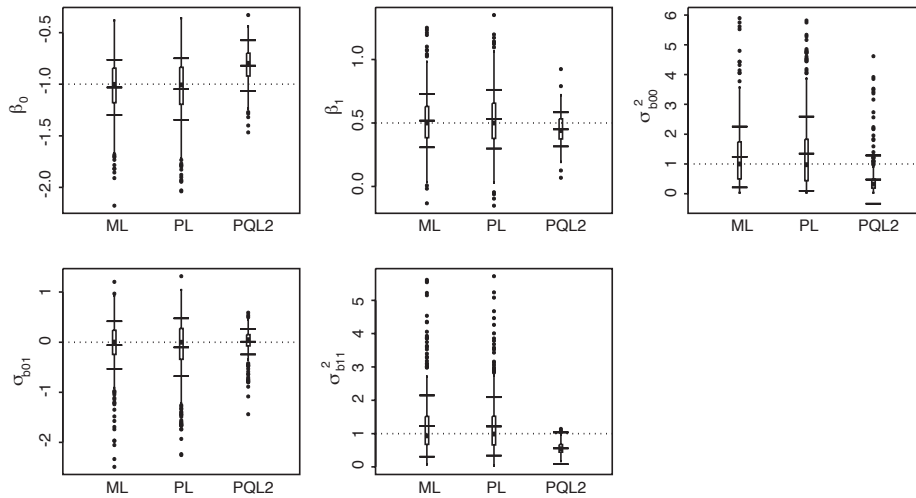


Fig. 6. Boxplots of ML, PL and PQL2 simulated parameter estimates under Model (10) with random slope.

PQL2 exhibits less variability, it seems to fail quite dramatically in this setting and variance parameters are seriously biased downwards.

4. Discussion

To conclude, we attempt to better delineate advantages and disadvantages of fitting multilevel models with binary responses using PL for estimation purposes.

One of the advantages of PL over ML stands at the computational level since the PL function involves evaluation of univariate and bivariate probits only, regardless of the number of specified random effects. Obviously, its use will not be very appealing when the number of random effects is small since ML estimation could be effectively used instead. It should be recalled that the computational cost to evaluate the marginal likelihood using a quadrature-based method increases exponentially with the dimension of integration. With PL, on the other hand, the complexity increases roughly as a quadratic function of cluster size.

To further illustrate this point, we report in Table 3 the median computing times for fitting model (9) to 100 simulated data sets for random effects of dimension 1–3. A quadratic term x_{ij}^2 was added to get random effects of dimension 3. As can be seen, PL is clearly not advantageous with random effects of dimension less than 3, at least in data sets with similarly large cluster sizes, but it becomes attractive with three-dimensional random effects.

Based on these considerations, we can tentatively recommend the use of PL estimation in multilevel probit models when:

- clusters are of small sizes, such as typically is the case in many longitudinal studies,
- and/or the random structure of the model to be fitted is complex.

Table 3

Median computing times (seconds) for fitting random-effects model (9) to 100 simulated data sets using ML and PL

	Dimension of random effects		
	1	2	3
ML	3.0	57.5	408.2
PL	86.4	116.0	185.8

Simulation settings:

- 20 clusters of size between 10 and 30
- random-effects variance equal 1

An additional advantage of PL is its numerical stability since it does not involve possibly unstable operations like matrix inversions. Convergence rates for PL were found comparable to those for ML in our simulation study. Approximate methods such as PQL, on the other hand, can sustain a great deal of numerical problems and from our personal experience, it is not uncommon that the algorithm fails to converge in practical applications, the problem being worsened as more complex models are fitted. It was clearly seen from our simulation study that the algorithm can behave well in some settings and fail miserably in others. Furthermore, the PQL2 algorithm appears to be sensitive to extreme success probabilities (i.e. ≈ 0 or ≈ 1). Thus, convergence rates in the first set of simulations were high because values chosen for the fixed parameters correspond to success probabilities in the mid-range (centered around 0.6) of the unit interval. However, they drop off significantly as soon as one departs from this setting. For instance, a value of $\beta_0 = -2$ in the random-intercept model (9) gives convergence rates of about 90% when $\sigma_{u0}^2 = 0.5$ and about 60% when $\sigma_{u0}^2 = 1$. This is even worse in the random-intercept-and-slope model, where our implementation of the procedure becomes highly unstable with convergence rates as low as 10%. If we add to this the fact that the PQL algorithm produces biased estimates and that the bias is dependent on the link specification and affects more or less severely different parameters, it becomes difficult to make definite recommendations about the use of this algorithm in general. It should be noted, however, that PQL runs much faster than either PL and ML so it might sometimes be the only procedure worth of consideration.

While PL estimation was advocated on the basis of its computational ease, this comes with a price, namely, loss of efficiency. Our simulation study showed a generally moderate loss (less than 20%), although it can attain up to 35% for some parameters.

Another drawback, clearly, is that PL will not be applicable to any hierarchically structured data set and practical limitations on the number of levels will arise. For instance, in a three-level model all possible pairs within and between level 2 units pertaining to the same level 3 unit should be considered. This will become computationally prohibitive as the number of levels and the cluster size increase. An application of PL estimation in a three-level model is described in [Renard et al. \(2002\)](#).

An interesting practical situation that can reduce the computational burden of the procedure is when covariates are discrete. In this case it suffices to consider solely

contributions of pairs among observations with different covariate values, while keeping track of the multiplicities for each combination of covariate values within a cluster. If this is coupled with the ordinary (unweighted) PL estimation procedure, then only observations with different covariate values across the whole data set need to be kept, thereby making evaluation of the PL function inexpensive. It should be observed that cluster membership information is still required to calculate the covariance matrix of the parameter estimates *via* the sandwich estimator.

In this paper we have focused exclusively on models dealing with a probit link specification. Even though a logit link would be straightforward to specify by assuming a standard logistic rather than normal distribution for ε_{ij} in (2), this yields intractable integrals for pairwise probabilities and thus, PL estimation cannot be applied in a beneficial manner. As discussed in Section 2, we do not see this as a strong limitation. One could also view the latent variable assumption as a restriction of the present approach. While this assumption may be sensible in many applications, it will not always be so. Moreover, the existence of the latent variable is usually unverifiable in practice. We emphasize, however, that this assumption was made only to facilitate mathematical developments (through probabilistic identities) and we do not necessarily need to believe that the threshold model holds.

Finally, when the response variable is measured on an ordinal rather than binary scale, [Hedeker and Gibbons \(1996\)](#) show how model (1) can be extended. In this case a number of thresholds or ‘cut-off’ points must be considered to define the different response categories, which then become additional parameters to be estimated in the model. The PL approach can be easily extended to deal with this kind of situations. Another area where PL methods might be useful is in crossed random-effects models where evaluation of the likelihood is particularly expensive with binary responses since high-dimensional integration is required.

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